From Hype to Health: Understanding Open Source Projects

Group 9 - OSS Pulse

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1. Motivation and Problem Statement

The AI open-source ecosystem is growing rapidly, especially in areas such as LLM tooling, agent frameworks, RAG systems, and model-serving infrastructure. However, high GitHub visibility does not always imply real adoption or long-term project health. Many repositories experience short-lived hype driven by social media or trend cycles, while other projects with lower star counts may have stronger practical adoption and healthier contributor communities.

This project proposes an end-to-end big data analytics platform that integrates GitHub activity data with Hugging Face model metadata to evaluate AI open-source projects from multiple perspectives: community health, adoption signals, development activity, and growth potential. The final system will be presented as a OSS analytics website with interactive charts and repository comparison views.

2. Target Users

This platform is designed for three primary user groups:

- **Open-source maintainers**, who want to understand whether their projects are healthy and whether their contributor communities are growing sustainably.
- **Developers and recruiters**, who want to identify which AI projects are genuinely impactful and worth adopting or following.
- **Researchers and investors**, who want to distinguish short-term hype from durable open-source momentum.

3. Core Research Questions

This project will answer the following questions:

1. Which AI repositories show strong real adoption versus short-term hype?
2. How are GitHub activity signals related to Hugging Face adoption signals?
3. Which repositories are under-recognized on GitHub but highly used in practice?
4. Which features best predict whether a repository will continue growing in the next 30 days?

5. What collaboration patterns exist among contributors in the AI open-source ecosystem?

4. Project Scope

To keep the project feasible within the course timeline, the scope will focus on a filtered set of AI-related repositories rather than the full GitHub universe.

- Analyze a list of approximately **100 AI/ML repositories**.
- Focus on repositories related to **agents, LLM tooling, RAG, diffusion, transformers, and model ecosystems**.
- Use one representative repository such as **OpenClaw** as a case study, but keep the main analysis at the ecosystem level.
- Prioritize **one year historical window** for GitHub activity data.

5. Data Sources

Data Source	Owner / Access	Estimated Size	Download Link	Team Responsibility
GH Archive (GitHub event stream)	Public GitHub activity archived by GH Archive; hourly JSON.gz files	~3–5 GB/day compressed; ~500 GB+ raw for 6 months before filtering; ~80–120 GB filtered Parquet	https://www.gharchive.org/	Jhe Chen Li
Hugging Face Hub model metadata	Public API via `huggingface_hub` / Hub API	~5–10 GB metadata snapshot	https://huggingface.co/docs/hub/api	Bo Yu
Optional: PyPI BigQuery download statistics	Public PyPI download and metadata tables in BigQuery	Varies by query; large public analytical tables	https://docs.pypi.org/api/bigquery/	Optional enrichment; team jointly profiles filtered subsets

6. Methods and System Design

6.1 Data Profiling

- Explore event type distributions, missing values, timestamp ranges, repository activity distribution, and author uniqueness.
- Profile Hugging Face metadata fields such as download distributions, tag coverage, and license types.
- The health scoring framework is anchored on the OpenSSF Criticality Score for measuring project importance. For repositories with Hugging Face presence, we further enrich the score with adoption signals including 30-day downloads and model ecosystem breadth.

6.2 Workflow Orchestration and ETL

- Use Airflow DAGs to schedule raw-data ingestion and upload tasks.
- Validate downloaded files before upload (file existence, size checks, basic schema checks).
- Parse nested GH Archive JSON into flattened structured tables.
- Extract GitHub repository links from Hugging Face model cards.
- Standardize task categories, licenses, and tags.
- Convert cleaned datasets into Parquet for downstream analytics.

6.3 Spark SQL Analytics

Core tables and views will include:

- `repo_daily_metrics`
- `hf_model_profiles`
- `repo_health_dashboard`
- `temporal_trends`

Planned analyses include:

- star/fork/contributor growth trends
- issue and PR responsiveness
- under-recognized repositories (high usage but low star count)
- ecosystem-level comparisons across task categories and frameworks

6.4 Structured Streaming

A streaming demo will replay hourly GH Archive files as a file stream to simulate near-real-time trend detection.

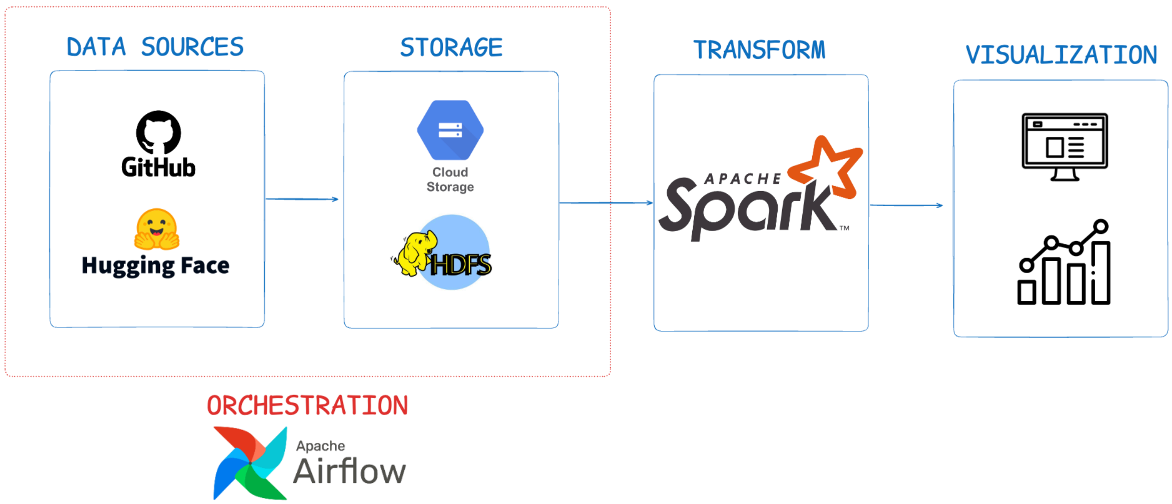
Example use cases:

- sudden star growth detection
- abnormal activity monitoring
- rolling 1-hour and 24-hour activity windows

6.5 Optional Graph Analysis

If time allows, GraphFrames will be used to build a contributor collaboration network and identify highly influential contributors or tightly connected communities.

Initial Design Diagrams



7. Frontend and Product Presentation

The final deliverable will be a lightweight analytics website.

Planned pages:

- **Overview page:** ecosystem summary, top repositories, fastest-growing projects
- **Repository detail page:** trend charts, health metrics, adoption signals
- **Comparison page:** side-by-side comparison across selected repositories
- **Insights/rankings page:** under-rated projects, healthy communities, growth predictions

8. Implementation Timeline

Week 1

- Finalize seed repository list
- Build Airflow Dags
- Upload raw data into HDFS
- Complete data profiling for both sources

Week 2

- Complete ETL and cleaning pipeline
- Build core Parquet tables
- Implement Spark SQL aggregate views

Week 3

- Build streaming demo
- Start dashboard/frontend implementation

Week 4

- Finalize visualizations and case study
- Polish website and prepare presentation
- Write final report